

08-Apr-2026

Agenda:

Starts -5: Sets

random variable

probability distribution

The language of outcomes: SETS

A set is a collection of distinct items.

set  $\rightarrow$  probability  
 $\downarrow$  values      what is the chance of  $x$  happening?  $\rightarrow$  prob

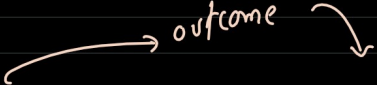


p.1: Every possible outcome of a dice roll

sample space ( $\Omega$ ) =  $\{1, 2, 3, 4, 5, 6\}$  (set)

Sets  $\rightarrow$  Events

Event A = "roll an even number" =  $\{2, 4, 6\}$  subset



Event B = "roll greater than 4" =  $\{5, 6\}$  subset

Event C = "roll a 9" =  $\{\}$  subset  
 $\downarrow$   
empty set  $\emptyset$

The four set operations

$$A = \{2, 4, 6\} \quad B = \{5, 6\} \quad \Omega = \{1, 2, 3, 4, 5, 6\}$$

$A \cup B \rightarrow$  union

$A \cap B \rightarrow$  intersection

$A^c \rightarrow$  complement

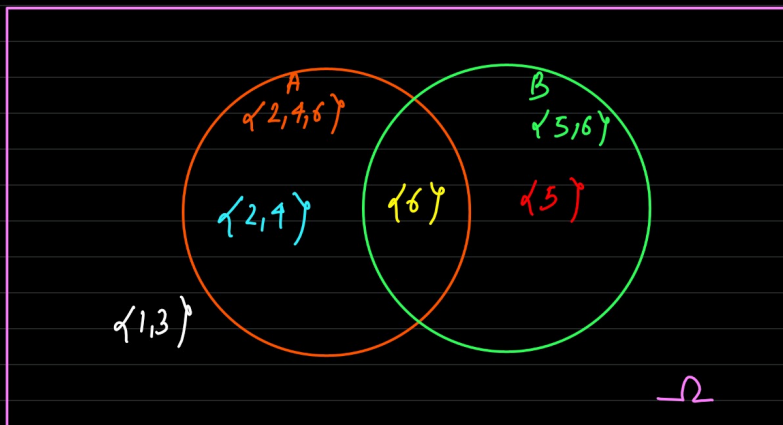
$A \setminus B \leftrightarrow A - B \rightarrow$  difference

$A \cup B \rightarrow A \text{ OR } B$

$U \rightarrow$  union

everything in A, everything in B, no duplicates

$$A \cup B = A + B - A \cap B$$



$$\begin{aligned} A \cup B &= A + B - A \cap B \\ &= \{2, 4, 6\} + \{5, \cancel{6}\} - \{\cancel{6}\} \\ A \cup B &= \{2, 4, 5, 6\} \end{aligned}$$

$$A \cap B \rightarrow A \text{ And } B \mid (A + B - A \cup B)$$

$\cap \rightarrow$  intersection

$\rightarrow$  values in both sets

$$A \cap B = A \text{ and } B$$

$$= \{2, 4, 6\} \text{ and } \{5, 6\}$$

$\downarrow$   
common

$$= \{6\}$$

$$A^c = \text{Not } A$$

$$= \Omega - A$$

$$= \{1, 2, 3, 4, 5, 6\} - \{2, 4, 6\}$$

$$= \{1, 3, 5\}$$

$$A \setminus B \mid A - B$$

$$= A \text{ but not } B$$

$\rightarrow$  elements in A that are not in B

$$A - B = \{2, 4, 6\} - \{5, 6\}$$

$$= \{2, 4\}$$

Random variable

Experiment: flip 2 coin

1 coin  $\rightarrow H, T$

2 coin  $\rightarrow HH, TT, HT, TH$

function (coin flip)  
 $\hookrightarrow$  number of heads  $\rightarrow$

$X \rightarrow$  "number of heads"  
 $\downarrow$   
number  $\rightarrow \underline{\underline{2, 1, 0}}$

outcome  $\rightarrow$  number

1 win 10 rupees if each head:

$$HH \rightarrow 20$$

$$TT \rightarrow 0$$

$$TH \rightarrow 10$$

$$HT \rightarrow 10$$

Random variable  $\rightarrow X$  is a function that converts outcome to a number

$$X(HH) \rightarrow 2$$

$$X(TT) \rightarrow 0$$

$$X(HT) \rightarrow 1$$

$$X(TH) \rightarrow 1$$

$X$  can take values from the set:  $\{0, 1, 2\}$

What is probability?

$\rightarrow$  How likely is an event?

Ex. 1 : Toss 1 coin

$$S = \{H, T\}$$

Event:

$$A: \{H\}$$

$$P(A) = \frac{\text{No. of favorable outcomes}}{\text{Total outcomes}} = \frac{1}{2} = 0.5$$

50% chance the event will be Head

Exp. 2: dice roll

$$S = \{1, 2, 3, 4, 5, 6\}$$

Event: Even number

$$A = \{2, 4, 6\}$$

$$p(A) = \frac{\cancel{1}, \cancel{4}, \cancel{6}}{1, 2, 3, 4, 5, 6} = \frac{3}{6} = \frac{1}{2} = 0.5$$

There is a 50% chance that if you roll a die you will get an even number.

Event: (not a even number)

$A \rightarrow$  even number

not a even number  $\rightarrow A^c$   
 $\rightarrow A'$

$p(A') \rightarrow$  not a even number

$$\boxed{\sum p(x) = 1}$$

$$p(A') = 1 - p(A)$$

$\downarrow$   
?

$$p(A') = 1 - \frac{1}{2} = \frac{2-1}{2} = \frac{1}{2} = 0.5$$

$$p(A') = 0.5 \rightarrow 50\%$$

$$3/6 + 3/6 \rightarrow 1$$

$$1, 3, 5 \quad 2, 4, 6$$

$$p(O) + p(E) = 1$$

$$S = \{H, T\}$$

$$p(H) = \frac{1}{2} \rightarrow \frac{1}{2} + \frac{1}{2} \rightarrow \frac{2}{2} \rightarrow 1$$
$$p(T) = \frac{1}{2} \rightarrow$$

$$\boxed{\frac{1}{2}} = \{ \text{Half filled, Half empty} \}$$

$$p(HF) + p(HE) \rightarrow 1$$

$$\begin{array}{c} \downarrow \\ ? \\ p(HF) = 1 - p(HE) \end{array}$$

$\downarrow$   
known

Solution 2 :

$P(A) \rightarrow A \rightarrow$  not a even number

$S: \{1, 2, 3, 4, 5, 6\}$

$A: \{1, 3, 5\}$

$$\frac{3}{6} \rightarrow \frac{1}{2} = 0.5$$

Experiment : OR condition

$S: \{1, 2, 3, 4, 5, 6\}$

Event:

$A: \text{even} \rightarrow \{2, 4, 6\}$

$B: \text{multiple of } 3 \rightarrow \{3, 6\}$

$P(A \cup B) \rightarrow$    $P$  prob of a number either even or multiple of 3

$$P(A) = \frac{3}{6}$$

$$P(A \cap B) = \frac{1}{6}$$

$$P(B) = \frac{2}{6}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{3}{6} + \frac{2}{6} - \frac{1}{6} = \frac{4}{6} = \frac{2}{3} = 66.67\%$$

↓  
chance that number is either even or multiple of 3

conditional probability:—

$$P(A|B)$$

prob of A given B  
already happened

$$P(\text{rain today} | \text{rained yesterday})$$

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$A: \text{even} : \{2, 4, 6\}$$

$$B: \text{multiple of 3} : \{3, 6\}$$

$$P(A|B)$$

prob of a even number when B happened

$$\{3, 6\}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

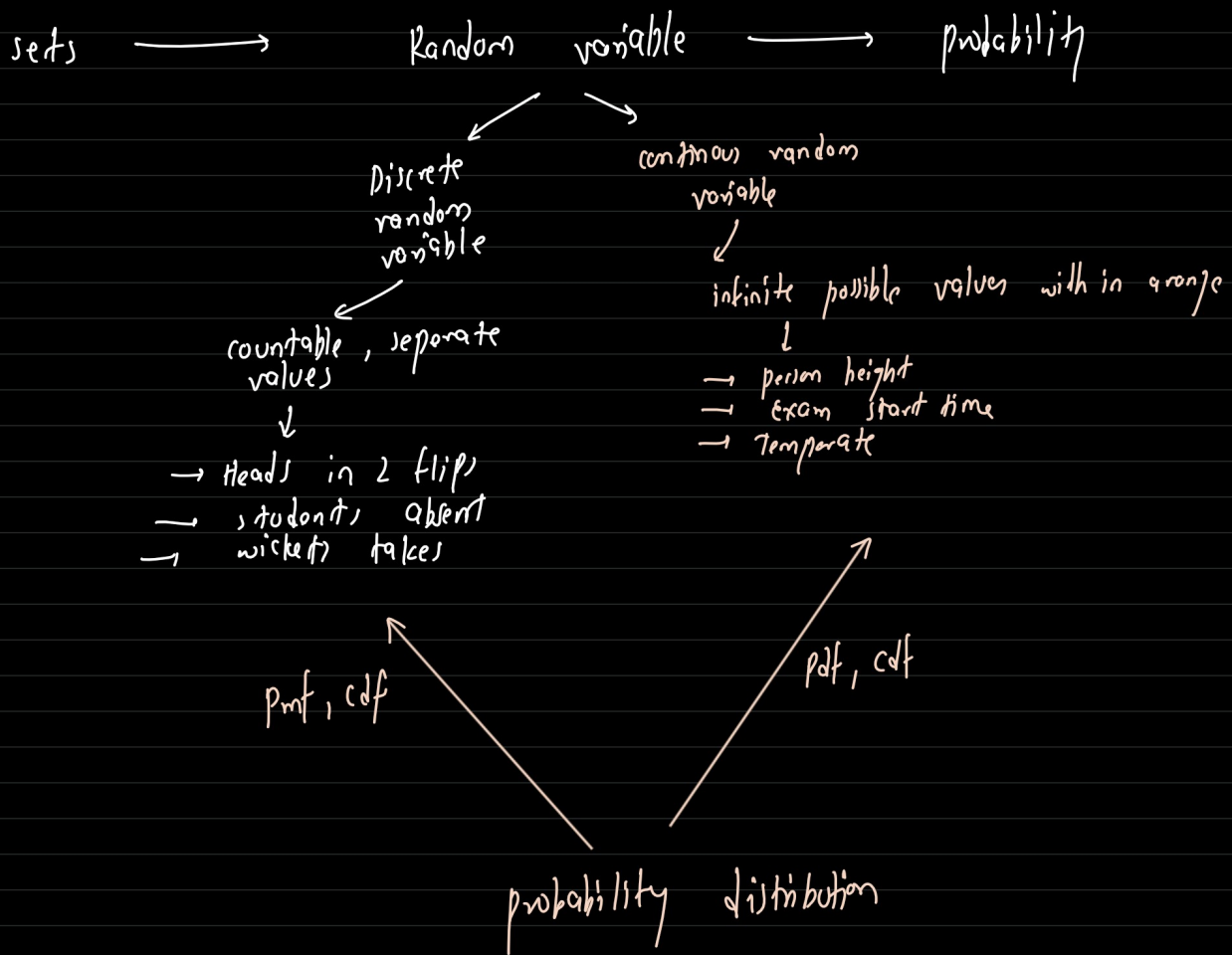
$$P(A \cap B) = \frac{1}{6} \rightarrow \{6\}$$

$$P(B) = \frac{2}{6} \rightarrow \{3, 6\}$$

$$P(A|B) = \frac{1/6}{2/6} = \frac{1}{2}$$

$$P(A|B) = P(A)$$

$P(B) \rightarrow \{3, 6\}$   
 $\swarrow$   
 $P(A)$   
 $\downarrow$   
 $\text{even} \rightarrow \frac{1}{2}$



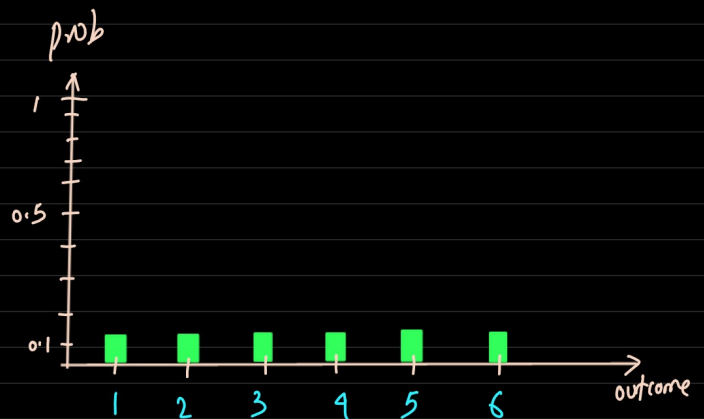
## pmf: Probability Mass Function

→ we map prob of each & every outcome

↳ what is the prob of value  $x$ ?

p.1 roll a dice

1  $\rightarrow 1/6$   
 2  $\rightarrow 1/6$   
 3  $\rightarrow 1/6$  ← map  
 4  $\rightarrow 1/6$   
 5  $\rightarrow 1/6$   
 6  $\rightarrow 1/6$





p.s: No of Heads in a 2 coin flip

TT  $\rightarrow$  0

HT  $\rightarrow$  1  $\rightarrow$

TH  $\rightarrow$  1

HH  $\rightarrow$  2

| x | count |
|---|-------|
| 0 | 1     |
| 1 | 2     |
| 2 | 1     |

$$p(x=0) = \frac{1}{4} = 0.25$$

$$p(x=1) = \frac{2}{4} = 0.5$$

$$p(x=2) = \frac{1}{4} = 0.25$$

